

FORT CHIPEWYAN, AB, Canada

WMO: 710560

Lat: 58.7664N

Lon: 111.1167W

Elev: 781

StdP: 14.29

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.3	-33.5	-46.3	0.4	-38.1	-41.5	0.5	-33.3	18.5	4.1	16.6	1.6	2.7	50	0.692

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	82.0	63.4	78.9	62.3	75.8	61.0	66.1	77.1	64.4	74.7	62.8	72.5	9.2	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.2	86.3	69.7	60.5	81.0	68.2	58.8	76.1	67.0	31.3	77.7	30.0	74.7	28.8	72.3	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.0	16.6	14.5	-44.5	87.9	4.3	3.0	-47.6	90.1	-50.1	91.8	-52.6	93.6	-55.7	95.8		
(5)			-44.0	69.0	3.9	2.2	-46.8	70.6	-49.1	71.9	-51.3	73.1	-54.2	74.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	30.2	-5.6	-1.0	11.1	30.5	46.1	58.3	63.3	59.2	48.7	34.4	14.6	0.2		
	DBStd	27.04	16.32	14.46	16.26	12.61	9.83	7.33	5.84	6.73	8.19	8.45	12.50	15.67		
	HDD50	8373	1724	1428	1205	588	189	17	0	9	120	487	1063	1543		
	HDD65	12808	2189	1848	1670	1034	588	220	101	199	490	948	1513	2008		
	CDD50	1134	0	0	0	3	69	267	414	296	81	4	0	0		
	CDD65	94	0	0	0	0	3	20	50	20	1	0	0	0		
	CDH74	1041	0	0	0	0	52	238	508	221	22	0	0	0		
	CDH80	184	0	0	0	0	8	40	95	38	3	0	0	0		
	WSAvg	6.6	5.7	6.1	6.7	7.8	8.1	7.3	6.7	6.4	6.5	6.3	6.0	5.8		
	PrecAvg	16.4	0.9	0.6	0.7	0.8	1.2	1.9	2.6	1.9	1.6	1.4	1.1	0.9		
(15) Precipitation	PrecMax	18.9	1.5	1.3	1.4	2.3	2.6	5.0	4.6	3.2	3.9	4.0	3.3	1.7		
	PrecMin	11.9	0.3	0.0	0.2	0.0	0.4	0.4	0.9	0.8	0.4	0.1	0.4	0.3		
	PrecStd	2.2	0.3	0.3	0.3	0.6	0.6	1.1	1.1	0.7	0.9	0.9	0.7	0.4		
	PrecStd	2.2	0.3	0.3	0.3	0.6	0.6	1.1	1.1	0.7	0.9	0.9	0.7	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.4	37.4	49.1	65.3	80.0	84.6	86.8	84.3	76.9	62.4	44.9	36.8		
		MCWB	31.8	33.7	38.8	48.7	57.7	63.4	66.1	65.8	61.4	51.3	39.1	32.9		
	2%	DB	28.7	29.6	42.9	58.6	73.6	79.9	82.6	79.5	70.6	55.4	37.9	30.5		
		MCWB	26.7	26.5	35.4	44.9	54.2	61.0	64.1	63.9	58.1	47.1	34.3	28.4		
	5%	DB	22.5	24.5	38.5	53.4	68.4	76.4	79.6	76.0	66.1	50.6	34.2	26.0		
		MCWB	21.3	22.3	33.1	42.1	52.0	59.8	63.4	62.1	55.8	43.9	32.1	24.4		
	10%	DB	17.1	19.7	33.9	48.6	63.6	72.9	76.4	72.6	62.1	46.7	31.4	21.5		
		MCWB	16.1	18.1	30.0	39.8	49.8	58.6	62.3	60.7	53.8	42.1	29.8	20.3		
	0.4%	WB	32.1	33.8	39.8	49.4	59.7	66.2	68.9	68.8	62.5	52.8	39.2	33.2		
		MCDB	34.3	37.5	48.3	63.1	75.8	79.0	81.5	79.8	74.8	60.3	44.1	36.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	26.9	26.9	36.0	45.4	55.8	63.8	66.6	65.6	59.3	47.5	34.9	28.3		
		MCDB	28.8	29.2	42.0	57.3	69.9	75.0	77.9	75.5	68.3	53.6	37.2	30.2		
	5%	WB	21.3	22.5	33.2	42.7	53.4	61.8	65.1	63.6	56.9	44.6	32.3	24.4		
		MCDB	22.6	24.6	38.3	52.3	66.5	72.4	75.5	72.8	63.6	49.6	34.0	25.8		
	10%	WB	16.2	18.2	30.0	39.9	50.8	59.9	63.6	61.8	54.9	42.3	29.8	20.3		
		MCDB	17.3	19.8	33.9	48.3	62.3	70.1	73.4	70.3	61.0	46.5	31.5	21.5		
	5% DB	MDBR	16.6	20.2	22.7	21.2	21.9	21.5	20.6	20.1	18.5	13.7	13.7	15.6		
		MCDBR	19.1	22.2	23.2	27.0	29.5	26.3	25.6	25.7	24.7	20.7	14.3	17.7		
		MCWBR	17.8	19.7	17.1	16.3	14.3	12.0	11.4	12.7	14.0	13.2	11.7	16.1		
		MCDBR	19.2	22.1	22.9	25.7	26.7	22.6	22.2	22.8	21.6	19.2	13.5	16.8		
	5% WB	MCWBR	17.9	19.9	17.2	16.0	13.7	10.8	10.8	12.1	12.7	12.7	11.3	15.6		
		MCWBR	17.9	19.9	17.2	16.0	13.7	10.8	10.8	12.1	12.7	12.7	11.3	15.6		
(40) Clear-Sky Solar Irradiance	taub		0.212	0.238	0.275	0.322	0.343	0.369	0.389	0.353	0.319	0.282	0.233	0.190		
	taud		2.226	2.318	2.296	2.362	2.424	2.356	2.366	2.441	2.547	2.555	2.289	2.122		
	Ebn at Noon		211	262	279	282	282	275	266	269	263	244	197	176		
	Edh at Noon		13	20	29	32	33	36	35	30	23	17	12	10		
(44) All-Sky Solar Radiation	RadAvg		152	430	944	1509	1836	1914	1759	1412	912	435	163	91		
	RadStd		18	32	51	94	156	133	106	79	82	49	21	8		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	+3.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page